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Software Requirement Specifications

Smart Attendance system using Face Recognition (Version 1.0)

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Summary

The **Face Recognition Attendance System** is a modern solution designed to automate attendance processes using facial recognition technology. Developed using Python and Django, the system captures and recognizes user faces in real-time using a webcam. This ensures accurate attendance marking, eliminates manual errors, and improves efficiency for institutions and organizations. The system supports key functionalities, including user registration, attendance management, reporting, and dashboard features. The use case diagram and process flow provide a clear representation of user interactions and system processes. Overall, this solution enhances attendance tracking through a secure, saleable, and user-friendly interface.

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1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the development of the Face Recognition Attendance System. The system will enable automatic attendance marking using face recognition technology, ensuring accuracy and efficiency in attendance management.

1.2 Scope

The Face Recognition Attendance System is designed to:

- Register users by capturing their images and storing them in a database.
- Capture images through a webcam and match them with stored records for attendance.
- Provide a dashboard for administrators to manage attendance records.

1.3 Project Overview

The system will utilize Python, Django framework, and the face_recognition package. Images will be stored securely, and attendance will be marked upon successful recognition. The system is saleable and suitable for educational institutions or organizations.

1.4 Objectives

- Automate the attendance process.
- Minimize errors in manual attendance systems.
- Provide a user-friendly interface for attendance management.
- Facilitate easy reporting and attendance tracking.

1.5 Target Audience

The primary users of this system include:

- Educational institutions
- Corporates requiring attendance tracking

1.6 Technologies Used

- **Programming Language:** Python
 - **Framework:** Django
 - **Packages:** face_recognition, Numpy
 - **Database:** SQLite or MySQL
 - **Frontend:** HTML, CSS
 - **Webcam Integration:** Real-time image capture for recognition.
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2. System Requirements

2.1 Hardware Requirements

- **Client Device:** Laptop/PC with a webcam
- **Server:** At least 8GB RAM, 4-core processor
- **Storage:** Minimum 10GB storage for database and image storage

2.2 Software Requirements

2.2.1 Functional Requirements

- User registration with image upload.
- Real-time face detection and recognition.
- Attendance record storage and retrieval.
- Attendance reporting and management for admins.
- Secure login and authentication for system users.

2.2.2 Non-Functional Requirements

- High accuracy in face recognition.
- Scalability for multiple concurrent users.
- Secure storage of user data and images.
- Fast response time for real-time face recognition.

2.3 Development Tools

- **Programming Language:** Python
 - **Framework:** Django
 - **IDE:** Visual Studio Code or PyCharm
 - **Packages:** Flask, Flask-MySQLdb, Numpy, face_recognition
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3. System Features

3.1 User Registration

- Users will register by uploading their images via the system.
- System will store user details (Name, ID) along with images in a secure database.

3.2 Attendance Marking

- System will use the webcam to capture live images of users.
- Captured images will be compared with stored user images for recognition.
- Attendance will be marked upon successful recognition.

3.3 Reporting and Dashboard

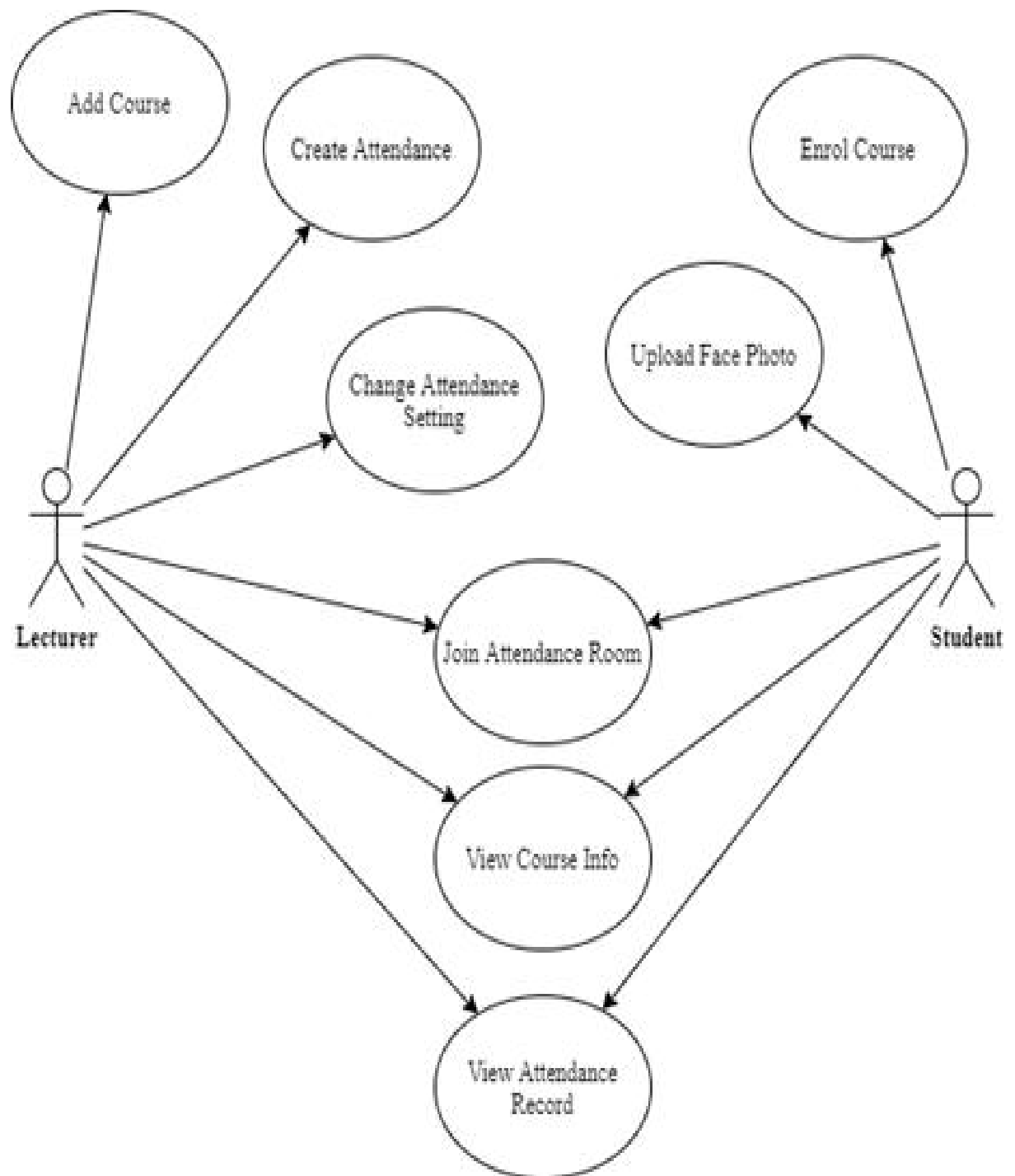
- Administrators can view attendance records and reports.
 - Reports can be generated for daily, weekly, and monthly attendance.
 - Admins can change attendance settings, add courses, and manage user data.
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4. Use Case Diagram

4.1 Purpose of Use Case Diagram

The use case diagram represents user interactions with the system, highlighting primary actions like registration, attendance marking, and report generation.

4.2 Use Case Diagram Representation



4.3 Use Case Descriptions

Use Case 1: Add Course

- **Actors:** Lecturer
- **Description:** Lecturer adds new courses to the system.

Use Case 2: Create Attendance

- **Actors:** Lecturer
- **Description:** Lecturer creates an attendance session for a course.

Use Case 3: Enroll Course

- **Actors:** Student
- **Description:** Students enroll in available courses.

Use Case 4: Upload Face Photo

- **Actors:** Student
- **Description:** Students upload their face photos for recognition during registration.

Use Case 5: Change Attendance Settings

- **Actors:** Lecturer
- **Description:** Lecturer updates or changes attendance settings.

Use Case 6: Join Attendance Room

- **Actors:** Student
- **Description:** Students join the attendance room for recognition.

Use Case 7: View Attendance Record

- **Actors:** Lecturer, Student
- **Description:** Users view attendance records and summaries.

Use Case 8: View Course Info

- **Actors:** Student
- **Description:** Students view details of the enrolled courses.

Use Case Table

Use Case ID	Use Case Name	Actor(s)	Description	Preconditions	Postconditions	Flow of Events
UC001	User Registration	User, Admin	This use case allows users to register their facial data into the system.	The user must provide valid information (name, image).	The user data is saved, and the system is updated.	1. The user inputs details (name, email, password). 2. The system validates the input. 3. The user uploads a face image. 4. The admin approves the registration.
UC002	Mark Attendance	User	This use case enables users to mark attendance using face recognition.	The user must be registered in the system.	Attendance is recorded in the database.	1. The user stands in front of the camera. 2. The system captures the image via webcam. 3. The system matches the face with the database. 4. The attendance is marked if a match is found.

UC003	View Attendance Report	Admin	This use case allows the admin to view and manage attendance reports.	Admin login must be authenticated.	Attendance reports are displayed.	1. The admin logs in. 2. The admin navigates to the attendance dashboard. 3. The system retrieves and displays attendance records.
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5. Process Flow

5.1 Process Flow Description

The process flow describes how the system handles user registration, attendance, and reporting:

1. Lecturer adds courses and configures attendance settings.
 2. Students enroll in courses and upload face photos during registration.
 3. During attendance, the webcam captures live images of users.
 4. The system matches the live image with stored records.
 5. Attendance is marked for successfully recognized users.
 6. Attendance records are stored and available for reporting.
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6. Conclusion

The Face Recognition Attendance System provides an efficient, accurate, and automated solution for attendance management. By utilizing Python, Django, and face recognition technology, the system ensures reliable attendance tracking while reducing manual errors. Future enhancements could include mobile app integration and advanced reporting features.

7. References

1. Python Documentation:

<https://www.python.org/>

2. Django Framework:

<https://www.djangoproject.com/>

3. Face Recognition Package:

<https://pypi.org/project/face-recognition/>

4. Numpy Library:

<https://numpy.org/>